Cover Letter

To the organizers of the Pacific Symposium on Biocomputing (PSB) 2027,

I submit the enclosed manuscript, "Three Routes to Failure: An Interpretable Diagnosis of Task Difficulty and Negative-Set Contamination in Genomic Sequence-Classification Benchmarks," for consideration.

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Requested session: "Biological Molecular Function: Beyond Homology"

This work fits the session's *beyond-homology* theme from the evaluation side: with interpretable, intervention-confirmed diagnostics, it shows that leading genomic foundation-model benchmarks are partly solved by homology leakage and compositional shortcuts rather than genuine molecular function—so moving beyond homology also requires benchmarks that do not reward those confounds.

Originality: This paper contains original, unpublished results and is not currently under consideration for publication elsewhere.

Authorship: I am the sole author of this paper.

Disclosure of LLM use (required): A large language model (LLM) was used as an AI assistant, under the author's direction, to help implement the experiments, run the analyses, and draft and revise the text. All quantitative results reported in the paper were produced by the released, deterministic, open-source code (fixed seed; results written to versioned CSV/JSON files) and were inspected and verified by the author. The author takes full responsibility for the correctness and content of the paper.

I intend to post a preprint of this manuscript to bioRxiv or arXiv.

Thank you for your consideration.

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Three Routes to Failure: An Interpretable Diagnosis of Task Difficulty and Negative-Set Contamination in Genomic Sequence-Classification Benchmarks

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Benchmarks that judge genomic foundation models (FMs) by their accuracy at classifying DNA sequences are increasingly relied upon, yet what makes a task "easy" or "hard" has been largely unexamined. We use an explicitly-interpreted, GPU-free (k-mer spectrum features and gradient-boosted trees) pipeline to diagnose performance across two benchmark suites (Genomic Benchmarks; the Nucleotide Transformer downstream tasks) and isolate three mechanistically-distinct routes to benchmark failure, each confirmed by intervention. First, motif *presence* does not predict success—across the nine Genomic Benchmarks tasks the number of enriched transcription-factor motifs is uncorrelated with learnability (Spearman -0.03)—whereas motif *discriminability* does: a semi-synthetic dose-response that dials the motif-only AUROC (d) of motifs implanted only in positives drives the benchmark MCC monotonically (Spearman $+0.924$, n=135; OLS slope $+1.99$, carried by three independent per-motif fits of $+1.96$ to $+2.02$). The real-task correlation corroborates this ($\rho = +0.74$, n=11, strengthening to $+0.87$ on the homology-clean subset), and out-of-sample a leave-one-task-out fit gives a stable mean absolute error of 0.14 MCC (with $R^2 = +0.44$ but a wide CI $[-0.27, +0.70]$ that includes 0 at n=11). Second, positional information governs learnability: position-binned k-mers give a *selective* rescue—one large recovery for the fixed-position task (splice sites, MCC $0.244 \rightarrow 0.868$), one marginal gain for the TATA-promoter, and $\sim$0 for the other twelve. Third, and most consequential, negative-set contamination: the negatives of two separately-curated human enhancer benchmarks carry an excess of TATA/AT-rich sequence (+20.9 to +22.7%). A GRCh38/GENCODE coordinate analysis refutes a core-promoter reading (flagged negatives are TSS-*depleted*); the excess is AT-rich background composition. We measure the artifact's magnitude directly—composition-equalizing the negatives collapses the `human_enhancers_cohn` k-mer MCC $0.461 \rightarrow 0.064$ and the TATA/GC motif-only AUROC to chance, and dinucleotide-shuffled negatives make it vanish. As an existence proof that pretrained models—not only our classifier—ride this shortcut, two architecturally-unrelated frozen genomic FMs (HyenaDNA; DNABERT-2) lose near-identical, significant enhancer AUROC when the composition artifact is removed, each against its own flat control. We frame the contamination as a *benchmark-design* artifact—the enhancer-vs-AT-rich-background composition gap is a real biological property exploited by a trivial axis, not dataset corruption—and release cleaned splits, the frozen-FM embedding cache (so the entire FM result reproduces with zero PyTorch), and a diagnostic command-line tool.

Keywords: interpretable machine learning; genomic benchmarks; regulatory sequence classification; negative-set contamination; k-mer features; DNA foundation models.

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1. Introduction

Benchmark leaderboards now drive much of the development of DNA foundation models (FMs)[2–6] — a model's standing is its aggregate accuracy across suites such as Genomic Benchmarks[1] and the Nucleotide Transformer downstream tasks[2] — yet the benchmarks themselves are treated largely as black boxes. We rarely ask *what property of a task makes it easy or hard*, or whether a high score reflects biological signal rather than an artifact of dataset construction. Without that, a leaderboard number is hard to interpret: two models with the same score may exploit very different things, and a "solved" task may be solved for the wrong reason.

We treat a simple, CPU-only classifier as an **interpretable diagnostic lens**, not a competitor to foundation models: frequency-normalized k-mer spectra classified with gradient-boosted trees (LightGBM[7]), so that — because k-mers map to short motifs — every prediction traces back to interpretable units. CPU-only is a property of the method, not a claimed contribution; we use frozen forward passes of pretrained FMs only as a probe (Section 4.5), neither fine-tuning them nor claiming to beat them.

Our contribution is a **mechanistic taxonomy of task difficulty** with three routes to failure, each demonstrated directly, replicated on an independent suite where possible, and confirmed by intervention:

(i) **Shared-motif** — the discriminative motifs are present in *both* classes, so they cannot separate them;
(ii) **Positional** — the signal is a fixed-position motif that a position-blind spectrum discards;
(iii) **Contamination** — the *negative* class is drawn from a different compositional distribution (an AT-rich background) that produces spurious separability, replicated across two human enhancer benchmarks.

We establish each route causally rather than by correlation or single example: (i) Route 1 by a dose-response at arbitrary sample size (n=135) plus a held-out test; (ii) Route 2 by a selective rescue across all 15 tasks and a bin-resolution sweep; (iii) Route 3's *magnitude* by composition-equalized cleaning and dinucleotide-shuffled negatives, and its *specificity* by an information-content-matched control-motif panel. As an existence proof that pretrained models ride the shortcut, two frozen genomic FMs lose significant enhancer AUROC when the artifact is removed, each against its own flat control.

A result ran *contrary to our initial expectation*: motif *presence* does not predict success — the hardest tasks are often the most motif-rich — whereas *discriminability* does. Throughout, "contamination" means a negative set carrying a label-correlated artifact: here an AT-rich compositional bias, not biological core-promoter leakage (which our Route 3 coordinate analysis tests directly). We release cleaned splits, the frozen-FM embedding cache, and a diagnostic tool so future evaluations can avoid the artifact.

2. Related Work

DNA foundation models and benchmark suites. Genomic FMs — DNABERT,[3] DNABERT-2,[4] the Nucleotide Transformer,[2] HyenaDNA,[5] Caduceus[6] — are pre-trained on large genomic corpora and ranked on downstream classification suites: Genomic Benchmarks[1] (nine datasets: promoters, enhancers, regulatory and open-chromatin regions, demonstration tasks) and the Nucleotide Transformer downstream tasks[2] (promoter, enhancer, splice-site, with published FM numbers). We do not propose or fine-tune an FM; we analyse these *benchmarks* and use frozen forward passes of two FMs (HyenaDNA, DNABERT-2) only as an interpretive probe (Section 4.5).

Negative-set composition bias in regulatory-sequence classification. That the *negative* set can dominate a regulatory-sequence classifier is a recurring, under-emphasized theme. The k-mer-SVM enhancer predictor of Lee, Karchin and Beer[19] and the gapped-k-mer SVM of Ghandi *et al.*[20] select features and negative controls to avoid trivial composition cues; their tooling generates GC-, length-, and repeat-matched negatives for exactly this reason.[21] Most directly, DeePromoter[22] foregrounds our Route 3: because TATA-like motifs are commoner genome-wide than in true promoters, models trained against easy genomic-background negatives exploit them, so the authors build harder, composition-matched negatives. Krützfeldt, Schubach and Kircher[23] study the same axis quantitatively, comparing GC-matched genomic negatives against k-mer-preserving shuffles — the two interventions we use — and find the negative-set choice strongly determines what a regulatory model learns. Our contribution is orthogonal: rather than choosing negatives to train a better predictor, we give a *coordinate-resolved, intervention-confirmed diagnosis* of negative-set composition contamination *on released benchmark splits* across two suites, with a cross-architecture frozen-FM probe and released cleaned splits. Inflation by negative-set construction has also been noted for enhancer–promoter interaction prediction.[15]

Non-foundation baselines. ConvNova[13] shows that from-scratch convolutional networks, without large-scale pretraining, are competitive with — and sometimes exceed — genomic FMs; a separate line uses embedding-plus-classifier baselines[14,24] to question how much pretraining adds. These establish *that* non-foundation methods are competitive; our contribution is different — we use a simple, interpretable model not to win but to *explain why* tasks are easy or hard (the three-route mechanism). We add no from-scratch CNN comparison: our cross-architecture evidence comes from frozen-FM probes and a model-family route-invariance analysis (Section 4.6).

3. Methods

The diagnostic pipeline is CPU-only (no GPU, no PyTorch/TensorFlow; see the FM-probe note below), uses a single fixed random seed (42) everywhere, and reports 95% percentile-bootstrap confidence intervals (CIs) with the individual test sequence as the resampling unit. Exact reproducibility parameters — software versions, LightGBM/LR hyperparameters, PSSM pseudocount and background, histogram-bin counts, bootstrap and permutation resample counts, and the pinned dataset revision — are listed in the supplement in the code repository (`https://github.com/rikhinkavuru/gb_kmer_benchmark`); everything needed to judge each procedure is below.

Featurization. Each sequence is its k-mer spectrum for $k \in \{3, 4, 5, 6\}$: counts over the fixed, data-independent vocabulary of all 4^k ACGT k-mers, L1-normalized to relative frequencies. The fixed vocabulary makes the feature space identical across datasets and splits (no vocabulary leakage); k-mers with non-ACGT characters are ignored. Matrices are sparse (CSR).

Models. The main model is gradient-boosted trees (LightGBM[7]) with hyperparameters fixed across all datasets; a scikit-learn[8] logistic-regression (LR) *floor* (with L2-normalized rows, since linear models are scale-sensitive) is also reported. Metrics are accuracy, the Matthews correlation coefficient (MCC)[10] and the area under the ROC curve (AUROC, one-vs-rest macro for multiclass); we emphasise MCC. Per dataset we select k for LightGBM on a held-out validation split — a stratified 80/20 carve of the training set — choose the k with the highest *validation* MCC, retrain on the full training split, and evaluate once on the test split, which is never used for model selection.

Per-TF motif enrichment. We score k-mers against the JASPAR 2024 CORE collection[9,12] as log-odds position-specific scoring matrices (PSSMs) and test, per TF, whether its motif score is *enriched* among high-gain k-mers via a gain-weighted statistic against an analytic permutation null, with Benjamini–Hochberg FDR[11] at $q < 0.05$. We avoid a "best match over the whole collection" threshold (with $\sim$880 profiles it *saturates*: 61–100% of k-mers are the optimal substring of *some* low-information window); the per-TF test does not saturate and names the responsible TF.

Motif-only discriminability. To ask whether motifs *separate the classes*, we scan each sequence (both strands) with a curated set of TF PSSMs, count hits above a fixed bits-per-position threshold, and compute the AUROC of motif-hit-count for the positive-vs-negative contrast ("motif-only AUROC"). The threshold is fixed across both classes, so any difference is signal, not a threshold artifact. For the GC-box (Sp1/KLF) we use 1.0 bits/pos; for TATA a specificity-controlled 1.3 bits/pos (TBP is a low-information AT-rich motif, so at 1.0 bits/pos hit counts track AT-content; a sweep shows the suspect-vs-control contrast persists at 1.3 while controls flatten, and the conclusion is intact across a 0.8–1.5 bits/pos sweep, Section 4.4). The same vertebrate TBP profile is used for all species including the `drosophila` control, since TBP and the TATA consensus are highly conserved.

Semi-synthetic discriminability dose-response (Route 1). To make the discriminability–learnability relationship causal at arbitrary n, we generate random ACGT backgrounds matched in length and base composition to a real task and implant PWM-sampled instances of a chosen JASPAR TF into positives (and not negatives) at a tunable rate, sweeping the rate to vary the realized motif-only AUROC d. Because the motif is implanted *only in positives*, this manipulates discriminability and is silent on motif *presence* (answered separately on real tasks, Section 4.2). For each setting we featurize, fit LightGBM, and measure test MCC; we repeat over five seeds and three motifs of differing information content (SP1, MYC, TBP). d is *measured*, not assumed, by the same scan used on real data.

Positional featurization. To test the positional route we add (alongside, not replacing) a position-binned spectrum: each sequence is split into B equal-fraction bins, the L1-normalized

k-mer spectrum is taken within each, and the bins are concatenated to the global spectrum. B=1 is the baseline; a sweep $B \in \{1, 2, 4, 8, 16\}$ measures how the rescue scales with bin resolution.

Contamination cleaning. We use three complementary cleanings of the *negative* set; positives are left identical. (i) *TATA-flag*: flag negatives carrying a high-confidence TATA box (TBP $\geq$ 1.3 bits/pos) and remove a seeded random subset equal to the excess (the neg-minus-pos TATA-box-rate gap $\times\, n_{\text{neg}}$, e.g. 22.2% of `cohn`'s negatives); since removal shrinks the negative set the cleaned AUROC moves toward but not fully to 0.5. (ii) *GC-matched*: subsample negatives to the positive-class GC histogram. (iii) *Fully composition-equalized*: we fit one logistic-regression propensity model $p = P(\text{positive} \mid \text{composition})$ on a 20-dimensional composition signature (4 mononucleotide + 16 dinucleotide L1-normalized frequencies), pooled over the train and test sequences, then resample negatives *within each split separately* so their propensity histogram matches the positives' — balancing GC and all 16 dinucleotides jointly (Rosenbaum–Rubin[16]), with balance verified by the per-feature standardized mean difference. Because equalized cleaning changes the test set the resulting MCC is *unpaired*; the motif/composition AUROC collapse is the causal readout.

Dinucleotide-shuffled controlled negatives (Route 3). For the `cohn` and `nt_enhancers` positives we construct matched negatives by an Altschul–Erikson dinucleotide-preserving shuffle[17] of the positives (Eulerian-path method; dinucleotide — hence mononucleotide and GC — composition is preserved exactly, verified per sequence). By construction these negatives have no composition gap, so any residual class signal is higher-order (motif) structure.

Negative-control-motif panel (Route 3 specificity). To test that the TATA skew is specific to the motif rather than manufactured by the scanning procedure, we build information-content-matched control PWMs: 20 column-permutations of the TBP PSSM (preserving per-column and total information content exactly while destroying the ordered TATA pattern), and 10 random PWMs matched to TBP's total information content. We scan the contaminated sets with each control using the *identical* both-strand, fixed-threshold, motif-only-AUROC procedure and compare to real TATA.

Coordinate analysis (Route 3). We recover GRCh38 coordinates for the negatives — by exact substring for `cohn` and by minimap2 alignment[18] for `nt_enhancers` — and intersect TATA-flagged versus non-flagged negatives with GENCODE v44 transcription start sites (TSS, $\pm$500 bp), reporting overlap rates, a Fisher exact test, and the TATA-to-nearest-TSS offset distribution.

Frozen FM-probe (existence proof). We extract frozen embeddings from two pre-trained genomic FMs — HyenaDNA-tiny (16k context, single-nucleotide tokenizer) and DNABERT-2-117M (512-token context, byte-pair-encoding tokenizer) — with CPU forward passes only (no fine-tuning, no GPU), mean-pooling the last hidden state over real tokens and additionally storing the sequence-summary token ([SEP] for HyenaDNA, [CLS] for DNABERT-2). On the frozen embeddings we train both an LR and a LightGBM head and evaluate with the *paired* protocol of Section 4.5. All torch usage is isolated to a separate embedding-extraction process (the diagnostic pipeline imports zero torch), and the embeddings are released as a cache so the entire downstream FM result reproduces with no torch and no model load. Significance for the motif-only AUROCs uses an empirical label-shuffle permutation null; the NT tasks are read at a pinned dataset revision (supplement).

4. Results

Table 1. Benchmark landscape across both suites: best-k LightGBM per task, ordered by MCC (descending). $n_{\text{train}}/n_{\text{test}}$ are the split sizes; CIs (95% bootstrap, test sequence the resampling unit) are on the test split. AUROC is one-vs-rest macro for the multiclass tasks (regulatory, enhancers-types, splice). The 'solved'/'composition' labels mark non-failing tasks; the three failure routes are shared-motif, positional, and contamination. `human_enhancers_ensembl` is labelled contamination (weak): its excess is only +10.8%, suggestive rather than robust. Every value is read directly from the result CSVs.

Suite	Task (dataset id)	n_{train}	n_{test}	k	MCC [95% CI]	AUROC	Route
GB	`demo_human_or_worm`	75,000	25,000	5	0.914 [0.909, 0.919]	0.992	composition
GB	`human_ensembl_regulatory`	231,348	57,713	5	0.907 [0.904, 0.910]	0.991	solved
NT	`nt_promoter_all`	53,276	5,920	4	0.888 [0.877, 0.899]	0.985	solved
NT	`nt_promoter_no_tata`	47,767	5,299	4	0.887 [0.873, 0.899]	0.985	solved
GB	`human_nontata_promoters`	27,097	9,034	6	0.874 [0.864, 0.883]	0.989	solved
NT	`nt_promoter_tata`	5,509	621	4	0.846 [0.803, 0.886]	0.984	solved
GB	`demo_coding_vs_intergenomic`	75,000	25,000	5	0.800 [0.793, 0.808]	0.965	composition
GB	`human_enhancers_ensembl`	123,872	30,970	6	0.593 [0.585, 0.602]	0.875	contam. (weak)
GB	`dummy_mouse_enhancers_ensembl`	968	242	6	0.529 [0.422, 0.632]	0.858	composition
NT	`nt_enhancers`	14,968	400	4	0.514 [0.432, 0.595]	0.850	contamination
NT	`nt_enhancers_types`	14,968	400	6	0.476 [0.415, 0.532]	0.837	shared-motif
GB	`human_enhancers_cohn`	20,843	6,948	5	0.461 [0.440, 0.482]	0.805	contamination
GB	`human_ocr_ensembl`	139,804	34,952	6	0.421 [0.412, 0.430]	0.785	shared-motif
GB	`drosophila_enhancers_stark`	5,184	1,730	4	0.398 [0.352, 0.441]	0.763	shared-motif
NT	`nt_splice_sites_all`	27,000	3,000	5	0.244 [0.219, 0.271]	0.704	positional

4.1. *Benchmark landscape*

Table 1 reports the best-k LightGBM per task with 95% bootstrap CIs; the discriminability–learnability relationship it summarises is established causally in Figure 1 (Route 1). Achievable MCC spans 0.244 to 0.914, and "solved" tasks are separated from "failed" ones by non-overlapping CIs. The same difficulty gradient appears in both suites: regulatory and promoter tasks are well-solved (`human_ensembl_regulatory`, `human_nontata_promoters`, the three NT promoters), whereas the enhancer and open-chromatin tasks are hard (`human_enhancers_cohn`, `human_ocr_ensembl`, `nt_enhancers`) and splice-site classification is poorly solved (`nt_splice_sites_all`) — all MCCs in Table 1. The taxonomy comprises three *failure* routes — shared-motif, positional, and contamination; the 'solved' and 'composition' labels mark non-failing tasks, not failure routes. These benchmark MCCs are seed-invariant (Section 4.6).

The absolute MCC of `human_nontata_promoters` is partly homology-inflated ($\sim$41% of its test sequences have a $\geq$0.7-Jaccard near-duplicate in training), inflating the Table 1 number only, not the within-split GC-box discriminability finding below. The broader audit (the contamination flagship `cohn` is redundancy-clean; `nt_enhancers` has verbatim overlap) is detailed in the Limitations.

4.2. *Route 1 — discriminability, not presence: a causal dose-response*

We hypothesized that well-solved tasks have discriminative k-mers matching known TF motifs. **Disconfirmed** — and the disconfirmation *is* the "presence does not predict success" half of Route 1: the number of FDR-enriched TF motifs does not track learnability (Spearman of MCC against #enriched TFs across the nine GB tasks $\rho = -0.03$, n.s.). The *failed* enhancer/OCR tasks are in fact the *most* motif-rich (`human_enhancers_ensembl` 265, `human_ocr_ensembl` 243 FDR-enriched TFs), while the solved promoter relies on far fewer families. (The dose-response below implants motifs only in positives, so it speaks only to discriminability, not presence; the presence claim rests solely on this real-task dissociation.)

What does predict learnability is motif **discriminability** — whether a motif is skewed to one class. In `human_nontata_promoters`, GC-box / Sp1–KLF motifs are concentrated in the positive (promoter) class: SP2 hit-count separates the classes with motif-only AUROC 0.751, and per-TF enrichment ranks SP1 first (gain-weighted $z = 7.60$, BH-FDR 1.4×10^{-11}), followed by SP2, SP4, KLF7, KLF12. In contrast, on the failed `human_ocr_ensembl` the E-box / AP-1 motifs that dominate its k-mer importance are present in *both* classes (TFAP4 E-box positive mean 0.103 vs negative mean 0.046, motif-only AUROC 0.527; BATF3 AP-1 0.648 vs 0.423, AUROC 0.573) — abundant but non-discriminative (the shared-motif route). The NT promoters replicate the GC-box-in-positives pattern (`nt_promoter_no_tata` SP1 hit-count positive mean 1.134 vs negative 0.428, AUROC 0.639).

The causal dose-response (Figure 1). The original evidence for "discriminability predicts learnability" was a pooled real-task correlation over n=11 tasks, underpowered within either suite. We now establish the relationship *causally* and at arbitrary sample size with a semi-synthetic dose-response (Methods): implanting a TF motif more in positives than negatives sets a target motif-only AUROC d, and we measure the resulting benchmark MCC. Across n=135 settings (3 motifs $\times$ 5 seeds $\times$ 9 doses), MCC rises monotonically with d (Spearman +0.924). Because these clustered settings are not independent, we do *not* lean on a pooled p-value; the load-bearing statistic is the agreement of *three independent per-motif slope fits*: +1.97 (SP1, 95% CI $[1.87, 2.07]$), +2.02 (MYC, $[1.90, 2.14]$), and +1.96 (TBP, $[1.78, 2.13]$). All three CIs exclude 0 and overlap each other, so the law is a property of task structure rather than of one motif's identity; the pooled OLS slope +1.99 (95% CI $[1.92, 2.06]$) is reported as an illustrative summary of the three. Higher-information motifs reach higher d and MCC. This synthetic, arbitrary-n causal result is the primary evidence for the discriminability half of Route 1.

Real-task corroboration and an out-of-sample test. The synthetic law is corroborated on the real tasks, and we lead with the homology-clean version. Dropping the two tasks whose absolute MCC is homology-inflated (`human_nontata_promoters`, `nt_enhancers`; Section 4.1) leaves nine TF-motif tasks on which discriminability predicts learnability strongly: Spearman $\rho = +0.87$ (p=0.003; partial controlling for suite +0.83; leave-one-out jackknife $[+0.81, +0.98]$, every subset positive and significant). On the full eleven-task set the correlation is $\rho = +0.74$ (p=0.01; GB-only +0.89; partial +0.75; jackknife $[+0.68, +0.83]$) — so removing the homology-inflated points *strengthens*, not weakens, the corroboration. Within either suite alone it is underpowered (n=5–6; NT-only $\rho = +0.40$, p=0.50), reported straight.

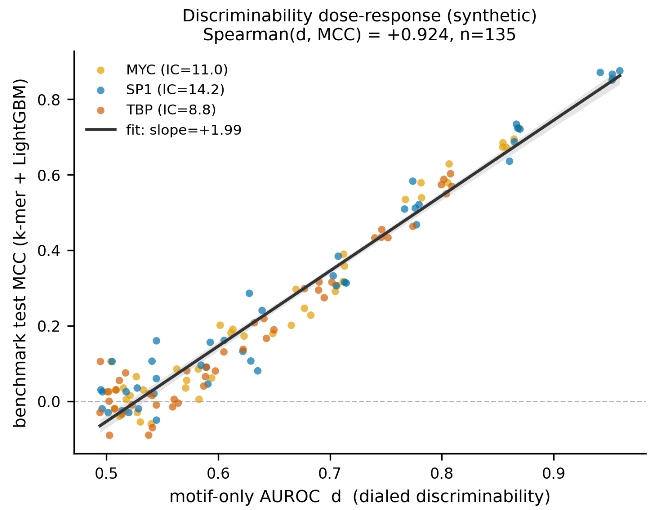

Fig. 1. Route 1 dose-response (semi-synthetic): benchmark MCC vs the dialed motif-only AUROC d for implanted TF motifs (n=135: 3 motifs $\times$ 5 seeds $\times$ 9 doses). MCC rises monotonically with d, and the per-motif slopes (SP1/MYC/TBP) overlap, so the law is motif-invariant (Section 4.2). Seed 42.

Out-of-sample, a leave-one-task-out fit predicts each held-out task with a stable mean absolute error of 0.14 MCC (95% CI $[0.11, 0.18]$); the corresponding $R^2 = +0.44$ has a wide CI $[-0.27, +0.70]$ that includes 0 at n=11, so we read it as corroborating, not a powered predictive claim, and lead with the MAE. (Excluded composition/non-TF tasks: human-vs-worm, coding-vs-intergenomic, `dummy_mouse_enhancers_ensembl`, `nt_splice_sites_all`; route assignment used the mechanism test, not correlation residuals, so the exclusion is not circular.)

4.3. *Route 2 — positional signal, selective and graded*

Splice sites fail on both suites (`nt_splice_sites_all` MCC 0.244; motif-only AUROC 0.522, at chance), consistent with a fixed-position GT–AG consensus the position-blind spectrum discards. Adding a position-binned spectrum (B=8) gives a clean **selective rescue** (paired bootstrap): splice recovers almost entirely, MCC $0.244 \to 0.868$ ($\Delta = +0.624$, 95% CI $[0.594, 0.651]$), while every control barely moves ($|\Delta| \leq 0.013$, $\sim$50$\times$ smaller). Extended to all 15 tasks the pattern is *selective* — one large rescue (splice $\Delta = +0.601$, $[0.572, 0.630]$, below the uncapped +0.624 because splice's training set is capped to 20k in this sweep), one marginal gain for the fixed-position TATA-promoter (`nt_promoter_tata` $\Delta = +0.019$, fixed -25 to -30 bp offset), and $|\Delta| \leq 0.028$ for the other twelve. Within the one responding task the effect is *graded* in bin resolution: $B \in \{2, 4, 8, 16\}$ grows the splice rescue monotonically $(+0.46, +0.53, +0.60, +0.66)$, not one lucky bin. Position is the causal missing ingredient exactly where fixed-position signal exists, and rescues neither motif-local nor contaminated tasks.

4.4. *Route 3 — negative-set contamination: existence, mechanism, magnitude, specificity*

Existence and replication. The strongest single-motif separator on the failed enhancer datasets is TATA/TBP, skewed to the **negative** class. On `human_enhancers_cohn`, negatives carry far more TATA boxes than positives (TBP hit-count positive mean 0.683 vs negative 1.641; motif-only AUROC 0.371, 95% CI [0.365, 0.376]; TATA-box fraction positive 34.6% vs negative 57.3%, an **excess of** +22.7%). The same signature appears on the separately-curated `nt_enhancers` (TBP positive 0.14 vs negative 0.58; AUROC 0.393 [0.387, 0.399]; TATA-box fraction positive 10.6% vs negative 31.5%, an excess of +20.9%) — so the contamination is **not GB-specific** but a broader benchmark-design issue. The negatives are systematically more AT-rich (lower GC and CpG O/E; `cohn`/`nt_enhancers` GC AUROC 0.737/0.822).

A benchmark-construction observation (not load-bearing). Exact full-sequence matching finds 100 of the 400 `nt_enhancers` test sequences (25.0%) are verbatim duplicates of a training sequence, which would inflate any model's *absolute* `nt_enhancers` MCC. It does not affect our within-split composition analyses (positives vs negatives inside one split), and is restated in the Limitations.

Mechanism: AT-rich background, not TSS leakage. Recovering GRCh38 coordinates for all 13,896 `cohn` negatives and intersecting with GENCODE v44 TSS **refutes** a core-promoter-leakage mechanism: TATA-flagged negatives overlap a TSS (±500 bp) in only 1.8% of cases, *below* the 3.3% of non-flagged negatives (0.55× depletion, Fisher $p=3\times10^{-8}$) and far below the 15.6% positive rate that validates the assay; flagged negatives near a TSS show no peak at the canonical −25 to −30 bp (median −96). The excess is thus a compositional artifact of AT-rich sampling, *not* core-promoter contamination. Extending to `nt_enhancers` by fuzzy minimap2 alignment (mappability 22%→28.2%), the flagged class is likewise TSS-*depleted* (0.6% vs 1.4%, 0.46×, mirroring `cohn`), no offset peak — **directionally corroborative but underpowered** (28% mappable; Fisher $p=0.17$).

Magnitude: how much of the enhancer signal is composition? The fully composition-equalized cleaning matches the negatives' joint GC + 16-dinucleotide distribution to the positives' (SMD 1.007 → 0.059 for `cohn`). On the equalized split the TATA and GC motif-only AUROC collapse to chance (`cohn` TATA 0.371 → 0.500, GC 0.737 → 0.500; `nt_enhancers` TATA 0.393 → 0.494, GC 0.822 → 0.540) and the k-mer MCC collapses onto a composition-only baseline (`cohn` 0.461 → 0.064, baseline 0.410 → 0.000). The GC-equalized collapse is *partly true by construction*, so confirmatory; the assumption-light controls below are the non-circular test. The clean control behaves oppositely: `drosophila` is already composition-balanced (the equalizer retains 42.9% of its negatives vs only 4.5–4.9% for the contaminated sets) and retains real signal above composition (MCC 0.398 → 0.177, residual +0.128). The equalized MCC is unpaired and underpowered, so the AUROC collapse is the causal readout. *A second, assumption-light intervention agrees*: against dinucleotide-shuffled negatives (no mono/dinucleotide composition gap by construction) the artifact *vanishes* for the contaminated sets (`cohn` TATA 0.371 → 0.551, GC → 0.500; `nt_enhancers` TATA 0.393 → 0.507, GC → 0.500), isolating the random-genomic sampling protocol as the cause; for the control, whose TATA is a genuine positive motif, the TATA AUROC instead *rises* to 0.749 (Figure 2).

Specificity: composition, not the ordered TATA pattern (Figure 3). A negative-control-motif panel shows the skew is not manufactured by the scanning procedure and is composition-driven. Information-content-matched *random* control PWMs produce no class

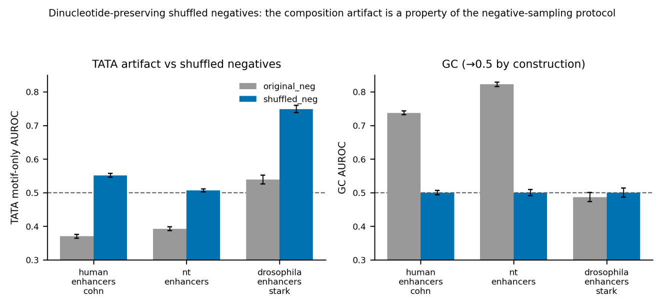

Fig. 2. Route 3 magnitude via dinucleotide-shuffled negatives. TATA (left) and GC (right) motif-only AUROC against real negatives vs dinucleotide-preserving shuffles of the positives. For the contaminated sets the artifact *vanishes* to chance once the composition gap is removed; for the clean control (`drosophila`), whose TATA is a genuine positive-class motif, it instead rises. Seed 42, 1000-bootstrap.

skew (motif-only AUROC 0.502 ± 0.033 for `cohn`, 0.507 ± 0.016 for `nt_enhancers` — on chance), confirming the scan invents no skew. But a *column-scrambled* TBP — which preserves TBP's AT-rich per-column composition while destroying the ordered TATA pattern — stays skewed (0.376 ± 0.017 and 0.396 ± 0.027, essentially equal to real TATA). The negative-set skew is therefore driven by AT-rich *composition*, not the ordered TATA pattern, corroborating the coordinate analysis. A built-in positive control behaves correctly: real TATA on `nt_promoter_tata` skews to the *positives* (AUROC 0.589, $[0.577, 0.600]$). The collapse is threshold-robust: sweeping the TATA threshold across 0.8–1.5 bits/pos, the equalized TATA AUROC stays on chance throughout (`cohn` $[0.485, 0.503]$, `nt_enhancers` $[0.490, 0.501]$) while the original is neg-skewed at every threshold — the headline does not depend on the 1.3 bits/pos point.

The interventional headline. Removing the excess TATA-bearing negatives (TATA-flag cleaning) drops 22.2% of `cohn` and 20.8% of `nt_enhancers` negatives and moves the TATA motif-only AUROC toward 0.5 (`cohn` $0.371 \to 0.435$; `nt_enhancers` $0.393 \to 0.485$; seed-invariant; GC-matched cleaning agrees, $\to$0.423/0.458). The apparent benchmark MCC also drops (`cohn` $0.461 \to 0.407$; `nt_enhancers` $0.514 \to 0.459$), consistent with partial inflation, though the MCC is unpaired so the AUROC collapse (and the paired FM `test_effect` below) is the causal readout. The *TATA-clean* control `drosophila_enhancers_stark` (a hard task with no negative-set TATA excess) behaves as required: the identical procedure removes 0.0% of its negatives and leaves its TATA AUROC (0.539) and MCC unchanged. We grade the verdict: `cohn` and `nt_enhancers` are robust; `human_enhancers_ensembl` is weaker/suggestive (excess +10.8%). Both robust benchmarks draw random AT-rich genomic negatives; the shuffled-negative result shows the artifact is a property of that sampling practice, not of either dataset.

4.5. *Cross-architecture frozen-FM probe: an existence proof*

The Route-3 results above are computed from the sequences themselves and so apply to any model, but they do not by themselves show that a *pretrained* model exploits the composition shortcut. We provide that as an existence proof with frozen-FM probes, and we are explicit that it is a **conservative lower bound**, not a measurement of the artifact's magnitude (which lives in Route 3).

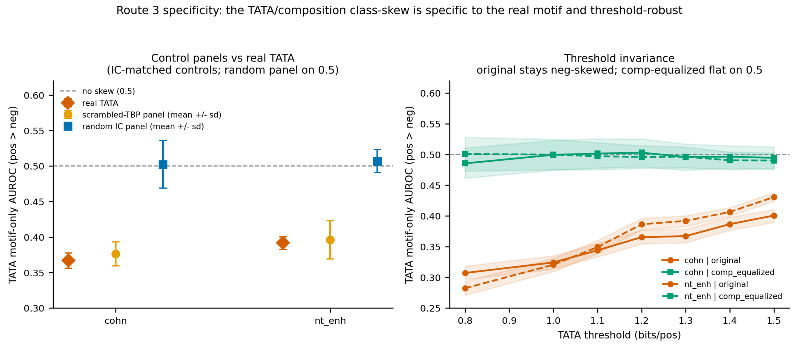

Fig. 3. Route 3 specificity. **(Left)** Random-IC control PWMs land on chance — the scan manufactures no skew — while column-scrambled-TBP PWMs (composition preserved, order destroyed) stay skewed ≈ real TATA: composition-driven, not order-driven. **(Right)** Across 0.8–1.5 bits/pos the composition-equalized TATA AUROC stays on chance, the original neg-skewed. Seed 42.

Paired evaluation. We train a head on a frozen FM's embeddings and measure how much enhancer AUROC the *same fixed head* loses when composition-biased negatives are removed from the *test* set (`test_effect`), on a retained GC-matched subset so the test set is held fixed (nested paired bootstrap, 1000 resamples); a flat control gives `test_effect` ≈ 0.

HyenaDNA. On HyenaDNA-tiny embeddings the fixed head loses significant enhancer AUROC when composition is removed: `test_effect` $= -0.048$ (95% CI $[-0.053, -0.043]$) on `cohn` and -0.079 ($[-0.107, -0.056]$) on `nt_enhancers` (both CIs exclude 0), while the clean control `drosophila` is flat, $+0.005$ ($[-0.003, +0.014]$, CI includes 0). The effect is **head-invariant** — an LR head gives the same signs and significance — so it lives in the frozen representation, not the read-out head.

DNABERT-2 (second architecture), with its own flat control. We repeat the identical paired evaluation with an architecturally-unrelated FM, DNABERT-2-117M (a bidirectional BPE BERT), on the two short datasets. DNABERT-2 loses near-identical AUROC: `test_effect` $= -0.047$ ($[-0.051, -0.042]$, mean-pool) and -0.046 ($[-0.052, -0.042]$, [CLS]) on `cohn`, and -0.072 ($[-0.100, -0.049]$) and -0.077 ($[-0.106, -0.051]$) on `nt_enhancers`; all CIs exclude 0, both poolings, both heads (Figure 4). DNABERT-2 cannot use the long `drosophila` control (its sequences exceed the 512-token context), so we give it *its own* flat control by a composition-representative placebo: on the same frozen `cohn` head we remove the *same number* of test negatives that the GC-matched cleaning removes (1148), but chosen at random (representative) rather than the AT-rich tail. This placebo is flat across all four head/pooling combinations (every `test_effect` CI includes 0), whereas removing the composition-biased negatives costs -0.047 — so DNABERT-2's loss is specific to *composition* removal, not to shrinking the negative set.[a]

Interpretation. Two architecturally-unrelated pretrained FMs lose nearly the same sig-

[a]The naive flat control — a dinucleotide-shuffled-negative version of `cohn` — is degenerate here: each shuffled negative has the same per-sequence GC as its positive, so GC-matched cleaning removes nothing and `test_effect` is trivially 0. The representative-removal placebo is the non-degenerate control.

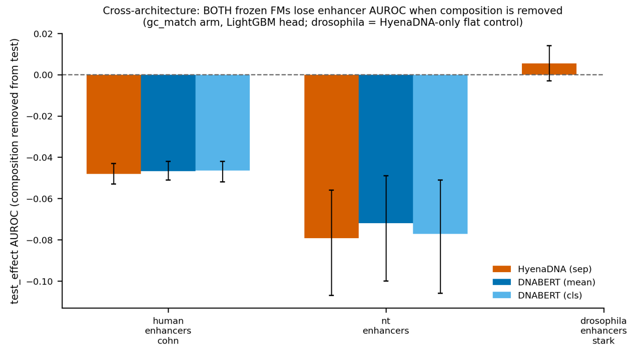

Fig. 4. Cross-architecture frozen-FM existence proof: the fixed FM head's enhancer AUROC loss (`test_effect`, GC-matched arm) when composition is removed, for HyenaDNA and DNABERT-2 (LightGBM head, 95% bootstrap CIs). Both lose near-identical significant AUROC on the contaminated sets (`cohn`, `nt_enhancers`); the HyenaDNA control (`drosophila`) and the DNABERT-2 placebo are flat. Seed 42.

nificant enhancer AUROC when the composition artifact is removed. That a pretrained model demonstrably rides the shortcut is the claim; the small AUROC delta is a *floor*, not the artifact's size (which is large; Section 4.4).

4.6. *Robustness*

The route assignments are stable under the choices a skeptic would question. A 5-seed LightGBM loop leaves the benchmark MCC seed-invariant (max SD 0.013 across eight key tasks), with the solved/hard tier stable for 7/8 (the lone flip, `nt_enhancers`, sits on the 0.5 cutoff). Across model families the per-task solved/hard tier agrees on 13/15 tasks for LR vs LightGBM and 3/3 for LightGBM vs the frozen FM-probe head; disagreements are capacity-driven or 0.5-boundary, not route changes, and the FM-probe independently agrees the enhancer tasks are hard — cross-family evidence without a from-scratch CNN. The FM composition-removal effect is head-invariant (Section 4.5) and the Route-1 slope is motif-invariant (Section 4.2).

5. Discussion

For benchmark design. The taxonomy gives a model-independent vocabulary for *why* a genomic task is easy or hard: a high score can reflect a clean class-specific motif (the promoter / GC-box case); a low score, diffuse or combinatorial signal, a fixed-position motif a representation discards, or — most consequentially — a negative set whose composition itself produces the discriminative signal.

The contamination implication: benchmark design, not dataset corruption. Two enhancer benchmarks reward separating enhancers from negatives partly on an excess TATA/AT-rich signal the coordinate analysis shows is background, not core-promoter leakage — available to any model, and ridden by our classifier and two frozen FMs alike. But enhancer positives *are* GC-rich relative to genomic background as a real biological property, so the composition gap is a genuine feature of the two classes, **not a corruption introduced in dataset construction**: the critique is one of *benchmark design* — the task is partly solvable by a trivial composition axis that does not test regulatory grammar. The interventions drive the artifact to chance while residual enhancer signal survives in the control, so the shortcut is separable from genuine signal. We fixed the flagging criterion, overlap test, and thresholds before obtaining the coordinates; the result refuted our initial core-promoter prior.

Limitations (reported straight). (i) Route 1 rests on the causal dose-response, with the real-task correlation and held-out test (Section 4.2) as corroboration only — the latter's R^2 CI includes 0 at n=11, so we lead with the stable MAE. (ii) The FM result is a frozen-feature existence proof (no fine-tuning, no leaderboard scores), so its effect on any specific fine-tuned FM score is unmeasured. (iii) The composition-equalized MCC is unpaired and underpowered, so the AUROC collapse is the causal readout; as the GC-driven collapse is partly true by construction, the dinucleotide-shuffle and column-scrambled-TBP controls are the primary non-circular specificity evidence (Section 4.4). (iv) We read each benchmark at its *provided* split without homology re-clustering: an 8-mer-Jaccard audit finds the flagship `cohn` redundancy-clean (0.11% $\geq$0.7 near-duplicate) but `human_nontata_promoters` (41% $\geq$0.7) and `nt_enhancers` (25% verbatim) inflated, so their *absolute* Table 1 MCCs are partly homology-inflated — shifting those numbers but not the route assignments, within-split interventions, or contamination magnitude, and the correlation strengthens when they are dropped (Section 4.2).

Artifact and reproducibility. We release cleaned train/test splits (`sequence,label`) for `human_enhancers_cohn`, `nt_enhancers`, and the `drosophila_enhancers_stark` control (TATA-flag, GC-matched, and composition-equalized variants), the frozen-FM embedding cache with manifest (regenerating the entire FM result with zero PyTorch; Section 3), and the `diagnose.py` CLI as a route-labeled report card for contamination-free enhancer evaluation (`https://github.com/rikhinkavuru/gb_kmer_benchmark`).

Future work. Fine-tuning foundation models on the original versus cleaned splits to measure the artifact's effect on FM scores directly; and extending the taxonomy with additional representations.

6. Conclusion

Treating a simple, interpretable, CPU-only classifier as a diagnostic lens, we give a three-route taxonomy — shared-motif, positional, contamination — for why genomic sequence-classification tasks fail, each confirmed causally: Route 1 by a semi-synthetic dose-response (Section 4.2) that corroborates out-of-sample and strengthens on the homology-clean subset; Route 2 by a selective, bin-graded positional rescue; and Route 3 by composition-equalized cleaning, dinucleotide-shuffled negatives, and a negative-control-motif panel showing two enhancer benchmarks carry an AT-rich *benchmark-design* confound in their negatives — not core-promoter leakage, and not dataset corruption — of large magnitude. As an existence proof, two architecturally-unrelated frozen FMs lose near-identical, significant enhancer AUROC when the artifact is removed, each against its own flat control. We release cleaned splits, the frozen-FM embedding cache, and a diagnostic tool for contamination-free evaluation.

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